

HYDRAULIC FRACTURING TREATMENT REPORT



Client	AGIBA
Well	17
Field	Raml
Formation	B III
Date	December 12th, 2011
Prepared by	Nelly Mohamed
Phone	+2 01220 455 999
E-Mail Address	<u>Nabbas5@slb.com</u>

Dear Sir,

Schlumberger would like to thank you for the opportunity to design and execute the hydraulic fracturing treatment of the well **Raml-17**. In the following report you will find the preliminary design, execution, and evaluation of the performed job. This information is important for further improvement and future reference.

On the first page you will find a Treatment Summary of the process, which is further described in the rest of the document.

Please do not hesitate to contact us if you have any inquiry regarding the content of this report or if you need to discuss any future job design.

Sincerely,

Nelly Mohamed

Production & Stimulation Engineer

Well Production Services – Schlumberger (DOWELL)

Egypt

I. TREATMENT SUMMARY

- A. The, **B III** formation of **Raml-17** was propped fractured on Decemeber 12th, 2011.

Summary of Main Treatment		
Parameter	#	Unit
Avg Treating Rate	28.8	Bpm
Avg Pressure	2000	Psi
Max Treating Pressure	4519	psi
HHP	1411.7	hhp
20/40 Carbo-Prop pumped	4752.1	lbs

- B. The following table summarizes the main fracture parameters:

Summary of Job Comparison		
Property	Design	Evaluation
PAD Dirty (%)	32.5	31
Total Proppant (lbs)	80000	4752.1
Maximum PPA	6	5
Fluid Efficiency (%)	28	30.3
Frac Xf (ft)	315.9	320.5
Frac height (ft)	62.9	71.9
Frac width (in)	0.196	0.147

Company: Agiba

Well: Raml 17

Formation: B III



FracCADE^{*} STIMULATION PROPOSAL

Operator : Agiba Petroelum Company
Well : Raml-17
Field : Raml Field
Formation : BIII

Well Location : Western Desert
City : Cairo
Country : Egypt

Prepared for	: Agiba Petroleum Company	Service Point	: EEG
Proposal No.	: 1	Business Phone	: 0
Date Prepared	: 12 /12/2011	FAX No.	: 02-3594526

Prepared by : Ahmed Samir Zaki
Phone : 0172-0001372
E-Mail Address : AZaki2@slb.com

* Mark of Schlumberger

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The Operator has superior knowledge of the well, the reservoir, the field and conditions affecting them. If the Operator is aware of any conditions whereby a neighboring well or wells might be affected by the treatment proposed herein it is the Operator's responsibility to notify the owner or owners of the well or wells accordingly.

Prices quoted are estimates only and are good for 30 days from the date of issue. Actual charges may vary depending upon time, equipment, and material ultimately required to perform these services.

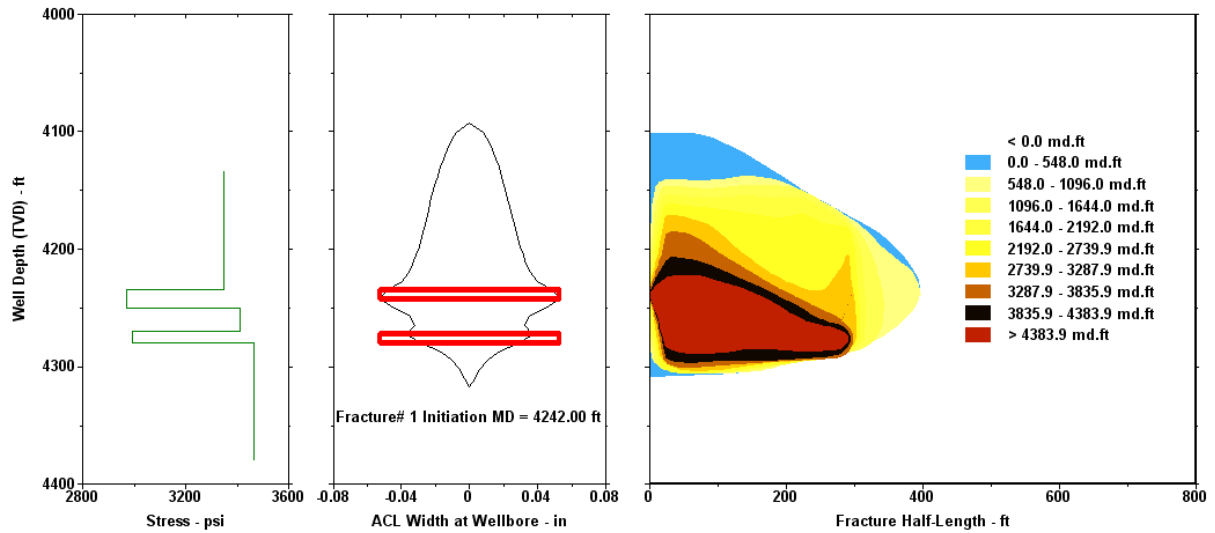
Freedom from infringement of patents of Schlumberger or others is not to be inferred.

Section 1: Propped Fracture Simulation Results

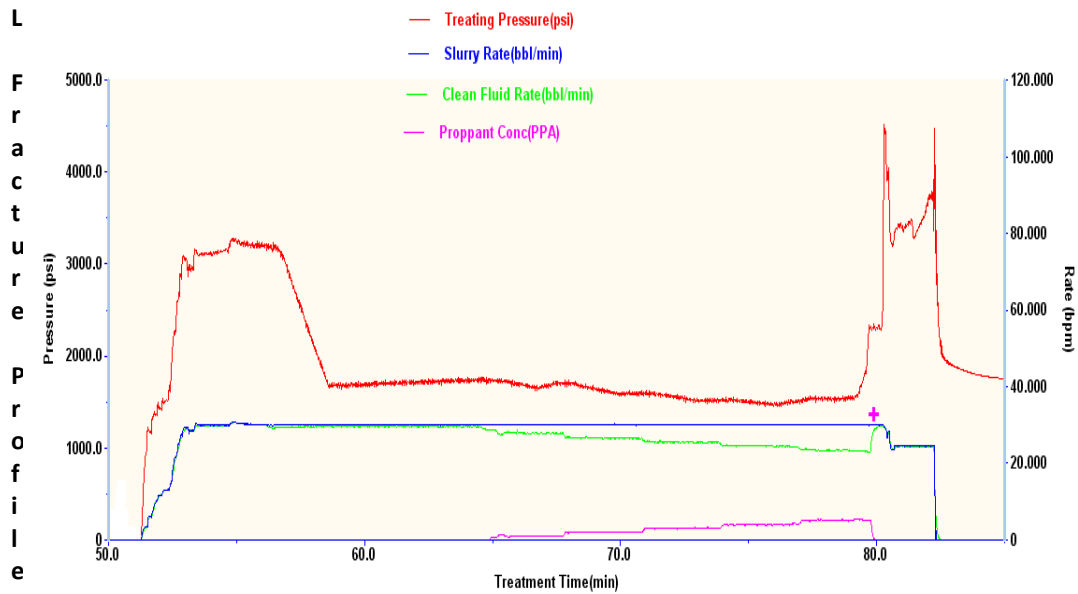
Agiba Petroleum Company
Raml-17

FracCADE*

ACL Fracture Profile and Conductivity



(1) A
C
L



and Proppant Concentration Plot

(2) Treating Plot

Section 2: Propped Fracture Simulation

The following are the results of the computer simulation of this Fracturing Proposal using a Pseudo 3-D Vertical model. Effective Conductivity and Effective Fcd are calculated based on perforated intervals with positive net heights.

Initial Fracture Top TVD..... 4234.0 ft
 Initial Fracture Bottom TVD..... 4250.0 ft

 Propped Fracture Half-Length..... 320.5 ft
 EOJ Hyd Height at Well..... 71.9 ft
 Average Propped Width..... 0.147 in
 Average Gel Fluid Retained Factor..... 1.00
 Net Pressure..... 1600 psi
 Efficiency..... 0.303
 Effective Conductivity..... 11009 md.ft
 Effective Fcd..... 8.1
 Max Surface Pressure..... 3425 psi

Simulation Results by Fracture Segment							
From (ft)	To (ft)	Prop. Conc. at End of Pumping (PPA)	Propped Width (in)	Propped Height (ft)	Frac. Prop. Conc. (lb/ft ²)	Frac. Gel Conc. (lb/mgal)	Fracture Conductivity (md.ft)
0.0	80.125	6.3	0.140	62	1.90	0.0	8423
80.125	160.25	7.2	0.173	82	1.73	0.0	10001
160.25	240.405	7.6	0.154	64	1.2	0.0	15201
240.405	320.5	1.1	0.12	42	0.5	0.0	7895

III. TREATMENT: AS MEASURED PUMPING SCHEDULE

As Measured Pump Schedule										
Stg #	Stage Name	Slurry Volume (bbl)	Slurry Rate (bbl/min)	Pump Time (min)	Fluid Name	Fluid Volume (gal)	Proppant Name	Max Prop Conc (PPA)	Prop Conc (PPA)	Prop Mass (lb)
1	Pre Flush	71.4	17.1	4.2	Pre-Flush	2982	None	0.0	0.0	0
2	Spacer	47.6	30.4	1.6	Brine 4% KCl	2000	None	0.0	0.0	0
3	PAD	261.9	30.0	8.7	ClearFRAC XT	11004	None	0.0	0.0	0
4	1.0 PPA	86.4	30.1	2.9	ClearFRAC XT	3491	S580 20/40	1.1	1.0	3491
5	2.0 PPA	89.4	30.0	3.0	ClearFRAC XT	3464	S580 20/40	2.2	2.0	6928
6	3.0 PPA	92.5	30.2	3.1	ClearFRAC XT	3470	S580 20/40	3.2	3.0	10411
7	4.0 PPA	95.5	30.0	3.2	ClearFRAC XT	3411	S580 20/40	4.1	4.0	13641
8	5.0 PPA	86.0	30.2	2.9	ClearFRAC XT	2818	S580 20/40	5.2	5.0	14226
10	Flush	59.1	24.1	2.6	Brine 4% KCl	2482	None	0.0	0.0	0

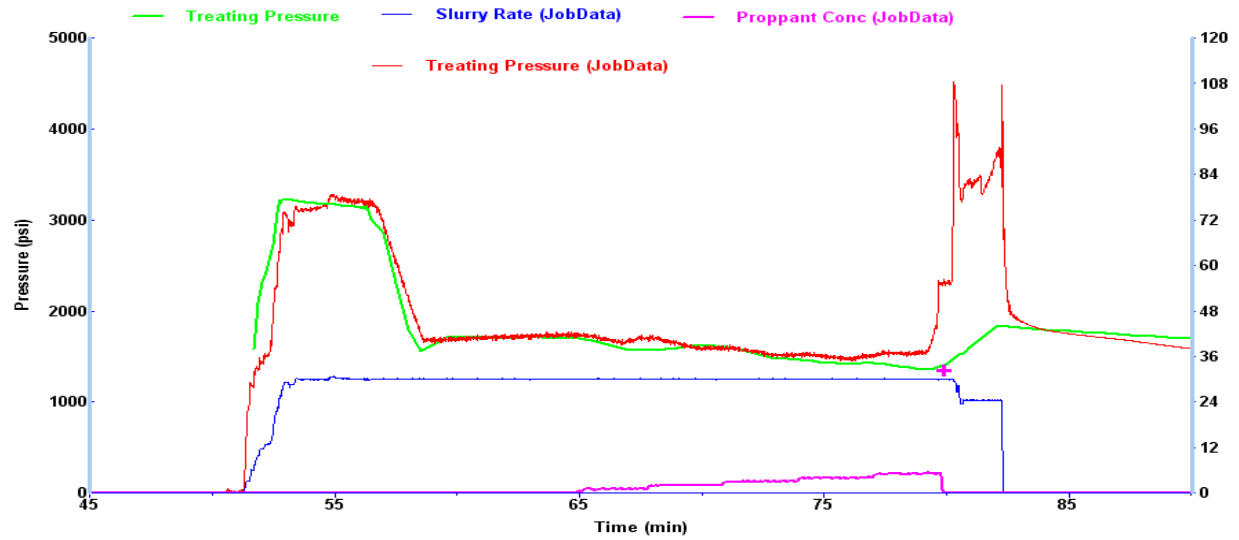
Stage Pressures & Rates						
Stg #	Stage Name	Average Slurry Rate (bbl/min)	Maximum Slurry Rate (bbl/min)	Average Treating Pressure (psi)	Maximum Treating Pressure (psi)	Minimum Treating Pressure (psi)
1	Pre Flush	17.1	30.4	1911	3160	19
2	Spacer	30.4	30.8	3217	3281	3121
3	PAD	30.0	30.2	1950	3238	1642
4	1.0 PPA	30.1	30.2	1696	1757	1624
5	2.0 PPA	30.0	30.3	1635	1733	1568
6	3.0 PPA	30.2	30.2	1550	1629	1486
7	4.0 PPA	30.0	30.2	1500	1547	1450
8	5.0 PPA	30.2	30.2	1622	2333	1495
10	Flush	24.1	30.1	3313	4519	0

As Measured Totals				
Pre-Flush (bbl)	Pump Time (min)	ClearFrac XT (gal)	Proppant (lb)	J - 456 (gal)
71.2	32.0	33600	47521	0.0

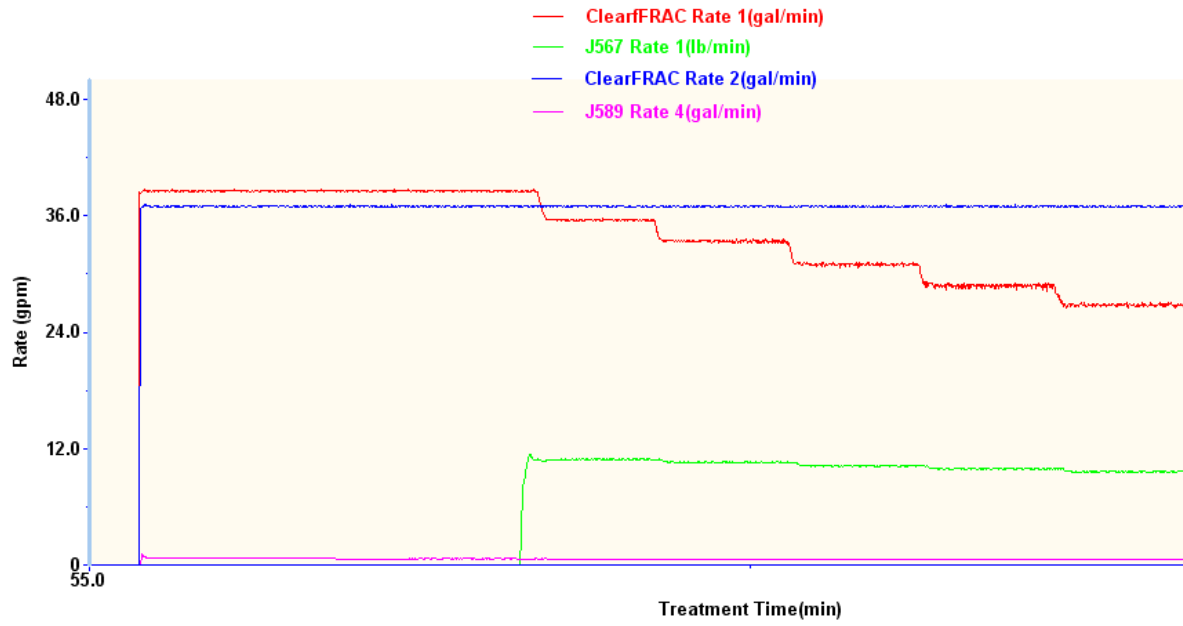
Average Treating Pressure: 2000 psi
 Maximum Treating Pressure: 4519 psi
 Minimum Treating Pressure: 0 psi
 Average Injection Rate: 28.8 bbl/min
 Maximum Injection Rate: 30.8 bbl/min
 Average Horsepower: 1411.7 hhp
 Maximum Horsepower: 3275.7 hhp
 Maximum Prop Concentration: 5.2 PPA

IV. MAIN TREATMENT: PRESSURE MATCH PLOT

Default Plot



V. MAIN TREATMENT: ADDITIVES PLOT



VI. MAIN TREATMENT: FRAC FLUIDS QA/QC



Fracturing Fluids QAQC Report

Client:	AGIBA Petroleum Company
Well:	RAML-17
Field:	RAML
Formation:	Bahareya III
BHST:	150 deg F
Date Prepared:	12-Dec-2011

Lab Engineer:	Haytham El-Kady
Phone No.:	+2 0122 7597694
E-Mail:	Haytham-elkady@hotmail.com

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SECTION 1: FLUID RECIPE

ClearFRAC XT:

Code	Concentration
M117, POTASSIUM CHLORIDE	334.00 lb/mgal
F105, EZEFL0(TM) Surfactant	100.00 gal/mgal
J590 ClearFRAC XT gelling agent	60 gal/mgal
J589 ClearFRAC XT Rheology modifier	0.5 gal/mgal
J567, Encapsulated Breaker	Variable ppt

SECTION 2: WELL SITE WATER ANALYSIS

Water Analysis Results										
Tank #	pH #	Temperature (degF)	Specific Gravity		Iron (mg/l)	Bicarbonates (mg/l)	Carbonates	Chloride Content		Hardness
			Pre KCl	Post KCl				Pre KCl	Post KCl	
1	7	66	1.000	1.019	<0.15	142	Nil	360	3.9%	250
2	7	64	0.999	1.022	<0.15	168	Nil	310	3.9%	250
3	7	65	1.001	1.022	<0.15	126	Nil	320	3.8%	250
4	7	65	0.999	1.024	<0.15	148	Nil	250	3.9%	250

SECTION 3: WELL SITE WELL SITE CLEARFRAC XT QA

- 1- ClearFRAC XT Shear recovery time = 6 seconds.
- 2- ClearFRAC XT Viscosity @511S⁻¹ @ 70 degF = 120 cP
ClearFRAC XT Viscosity @511S⁻¹ @ 110 degF = 90 cP